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Applied Analytics Project - Week 12 Assignment

The final trained model was saved as a .pkl file using Python's pickle library.

Environment Dependencies:

Operating System: The project was developed in a macOS environment.

Python Version: The notebook was developed using Python 3.10.

Python Packages and Versions: The main packages used in the notebook are: pandas (2.2.3), numpy (2.1.3), scikit-learn (1.5.2), pickle (built-in), pathlib (built-in).

Deployment Strategy:

For this project, the model was deployed using batch inference during the live 2026 NCAA March Madness tournament. Batch inference made the most sense because tournament games happen in scheduled rounds, so predictions only needed to be generated before each round started.

Once our final model was completed, we used it to predict the outcomes of the real March Madness games as the tournament was happening live. Before the first round began, the model generated predictions for every matchup using the pre-tournament team statistics. After the games finished, we compared the model's predictions to the actual game results to measure performance. We repeated this process for each round of the tournament which allowed us to test the model on real-world unseen data throughout the tournament.

We chose batch deployment instead of real-time inference because predictions did not need to be updated continuously during games. Generating predictions before each round was simpler and still allowed us to evaluate the model in a realistic live environment.

Monitoring Performance Metrics and Thresholds:

Model Accuracy: Model accuracy will be tracked weekly by comparing predictions against actual game outcomes. This metric is important because it provides a direct measurement of how often the model correctly predicts game winners.

- **Green**: Accuracy drop less than 3% from baseline performance
- **Yellow**: Accuracy drop between 3% and 7%
- **Red**: Accuracy drop greater than 7%

Prediction Positive Rate: The percentage of games predicted as wins for Team 1 will also be monitored. This metric is important because sudden changes in prediction behavior may indicate unstable model outputs or shifts in the underlying dataset.

- **Green**: Change less than 5% from historical prediction rates
- **Yellow**: Change between 5% and 10%
- **Red**: Change greater than 10%

Population Stability Index (PSI): The notebook calculates Population Stability Index values for features between baseline data and recent production data. This metric is important because it helps identify data drift. If feature distributions begin to shift significantly compared to the training data, the model may eventually perform worse in production.

- **Green**: PSI less than 0.10
- **Yellow**: PSI between 0.10 and 0.20
- **Red**: PSI greater than 0.20

Business Metric (ROI): Return on investment (ROI) from betting on the teams predicted to win by the model will also be monitored. This metric is important because the overall goal of the project is not only to predict game outcomes accurately, but also to determine whether the model can generate positive economic value against betting markets. By tracking ROI, we can evaluate whether the model's predictions are actually profitable over time.

- **Green:** Positive ROI over the tournament sample
- **Yellow:** ROI near 0% or slightly negative
- **Red:** Consistently negative ROI across multiple tournament rounds or seasons

Risk Mitigation Strategies:

Green Status: If all metrics remain within the green threshold ranges, the model will remain in production under normal monitoring procedures. Weekly reports would be reviewed to confirm that the model is still performing as expected.

Yellow Status: If one or more metrics enter the yellow threshold range, the model will remain active, but additional monitoring and investigation will be performed. The team would begin analyzing which metrics are declining and determine whether the issue is temporary or the beginning of a larger problem.

Red Status: If metrics enter the red threshold range, the model would be temporarily shut down because the predictions may no longer be reliable. A full investigation would be performed to identify the source of the issue. Before redeployment, the updated model would need to pass validation testing to confirm that performance metrics are back to acceptable levels.

Model Retrain Frequency:

The model would be retrained yearly after each March Madness tournament season is completed. Retraining once per tournament season makes sense because new team statistics, play styles, coaching, and roster changes can change the relationships between the features and game outcomes over time. By retraining the model with the newest tournament data each year, the model can better adapt to recent trends and reduce the risk of concept drift.

Data Drift and Concept Drift:

This project can be affected by both data drift and concept drift over time. College basketball changes from season to season, so the data and patterns the model learned from older tournaments may not always match newer tournaments.

In the Week 12 notebook, we checked for data drift using PSI scores between the training data and newer production data. The PSI values were all below 0.10, which suggests that there is currently little evidence of significant data drift in the feature distributions.

However, the notebook also showed that the model performed worse on newer tournament data compared to older data. This suggests possible concept drift where the patterns tied to winning games may be changing over time.

To reduce these risks, we would continue monitoring PSI scores, model accuracy, and ROI throughout the lifecycle of the model. The model would also be retrained after each March Madness tournament season using newer data so it can adjust to recent trends and tournament patterns.